

Olympiad Test Paper — Mathematics (Class 7) — Paper 5

Chapter 5: Parallel and Intersecting Lines

Paper 5 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	5 — Parallel and Intersecting Lines
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Every item chains four or more relations, so a single slip lands you on a pre-computed wrong option: equating co-interior angles instead of supplementing them (they add to 180 degrees, they are NOT equal); treating a non-parallel crossing as parallel; applying a corresponding or alternate equality before the lines are known to be parallel; halting one linear-pair or one bisector step early; using 360 degrees where 180 degrees applies (or the reverse); and forgetting that only a perpendicular crossing yields four equal angles. Do not write on this paper — mark answers on your answer sheet.

1. Two lines AB and CD cross at O. The angle $\text{AOC} = (3x + 14)^\circ$ and its adjacent angle $\text{AOD} = (2x + 16)^\circ$ form a linear pair. Ray OE bisects angle AOC. Find angle EOC.

- (A) 104 degrees (B) 38 degrees
(C) 52 degrees (D) 26 degrees

2. Two lines intersect at O so that one of the four angles is $(4x - 6)^\circ$ and the angle vertically opposite it is $(2x + 30)^\circ$. Find the measure of the angle adjacent to these (its linear-pair neighbour).

- (A) 66 degrees (B) 18 degrees
(C) 114 degrees (D) 33 degrees

3. Two lines intersect at O. Reading once around, the four angles are $(2x)^\circ$, $(x + 30)^\circ$, $(2x)^\circ$, $(x + 30)^\circ$. After finding x, ray OF bisects one of the $(2x)^\circ$ angles. Find the angle between OF and the nearer arm of that bisected angle.

- (A) 100 degrees (B) 40 degrees
(C) 80 degrees (D) 50 degrees

4. At the crossing of two lines, an angle is $(5x + 8)^\circ$ and its adjacent linear-pair angle is $(4x + 10)^\circ$. After finding the two angles, find the positive difference between the larger angle and HALF of the smaller angle.

- (A) 16 degrees (B) 98 degrees
(C) 41 degrees (D) 57 degrees

5. Two lines meet at O. One angle is $(7x - 5)^\circ$ and the angle vertically opposite to it is $(3x + 47)^\circ$. Find x.

- (A) 23 (B) 13
(C) 26 (D) 6.5

6. Three rays from O divide the full turn into angles $(2y)^\circ$, $(3y + 20)^\circ$, and $(4y - 20)^\circ$. After finding y, find the SUPPLEMENT of the largest of the three angles.

- (A) 140 degrees (B) 40 degrees
(C) 100 degrees (D) 20 degrees

7. A transversal cuts two parallel lines p and q. A corresponding angle at p is $(5x - 12)^\circ$ and the matching one at q is $(3x + 8)^\circ$. After finding x, the corresponding angle at q is bisected; find the acute angle the bisector makes with the transversal.

- (A) 38 degrees (B) 71 degrees
(C) 76 degrees (D) 19 degrees

8. A transversal cuts two parallel lines m and n. A pair of alternate interior angles measure $(3x - 5)^\circ$ and $(x + 41)^\circ$. After finding x, find the co-interior partner of that angle at the SAME crossing.

- (A) 64 degrees (B) 116 degrees
(C) 23 degrees (D) 46 degrees

9. Two lines AB and CD intersect at O. Angle AOC = 70° . Ray OE bisects the vertically opposite angle BOD. Find the angle between OE and OB.

- (A) 35 degrees (B) 70 degrees
(C) 55 degrees (D) 110 degrees

10. Two lines intersect at O. Angle AOC = $(6x + 4)^\circ$ and its vertically opposite angle is $(8x - 30)^\circ$. Find the linear-pair neighbour of angle AOC.

- (A) 106 degrees (B) 17 degrees
(C) 74 degrees (D) 53 degrees

11. Four rays from O make four consecutive angles around the point: a° , $(a + 18)^\circ$, $(a + 36)^\circ$, $(a + 54)^\circ$. After finding a, find the difference between the largest angle and the smallest angle.

- (A) 18 degrees (B) 54 degrees
(C) 36 degrees (D) 108 degrees

12. Three rays from O split the full turn into angles in the ratio 2 : 3 : 5. Find the SUPPLEMENT of the largest of the three angles.

- (A) 30 degrees
- (B) 60 degrees
- (C) 90 degrees
- (D) 0 degrees

13. Angles around point O are $(3x)^\circ$, $(2x + 10)^\circ$, $(x + 20)^\circ$, and $(4x - 30)^\circ$, in order. After finding x, find HALF of the largest of the four angles.

- (A) 114 degrees
- (B) 54 degrees
- (C) 36 degrees
- (D) 57 degrees

14. Two intersecting lines make four angles; three of them, read around, are $(2x + 10)^\circ$, $(3x)^\circ$, and $(2x + 10)^\circ$. After finding x, find the supplement of the $(3x)^\circ$ angle.

- (A) 102 degrees
- (B) 78 degrees
- (C) 60 degrees
- (D) 120 degrees

15. A transversal cuts parallel lines l and m. Co-interior angles measure $(4x + 10)^\circ$ and $(5x - 1)^\circ$. After finding x, find the alternate interior angle of the LARGER co-interior angle.

- (A) 86 degrees
- (B) 94 degrees
- (C) 90 degrees
- (D) 85 degrees

16. A transversal cuts parallel lines a and b. A corresponding angle at a is $(7x - 20)^\circ$ and at b is $(4x + 16)^\circ$. After finding x, find the co-interior partner of that angle at the OTHER crossing.

- (A) 116 degrees
- (B) 64 degrees
- (C) 12 degrees
- (D) 44 degrees

17. A transversal cuts parallel lines p and q. An exterior angle on the upper line is 137° . Find the co-interior interior angle on the lower line that lies on the SAME side of the transversal as that exterior angle's own interior linear-pair partner.

- (A) 43 degrees
- (B) 137 degrees
- (C) 90 degrees
- (D) 47 degrees

18. Five rays from O divide the full turn into angles $(y)^\circ$, $(2y)^\circ$, $(3y)^\circ$, $(4y)^\circ$, $(5y)^\circ$. After finding y, find the difference between the largest angle and the supplement of the second-smallest angle.

- (A) 12 degrees
- (B) 108 degrees
- (C) 132 degrees
- (D) 60 degrees

19. Lines l and m are parallel, cut by transversal t. Angle 1 (interior at l) corresponds to angle 2 at m; angle 2 is vertically opposite angle 3; angle 3 forms a linear pair with angle 4. If angle 1 = 121° , find angle 4.

- (A) 121 degrees
- (B) 90 degrees
- (C) 29 degrees
- (D) 59 degrees

20. Parallel lines l and m are cut by a transversal. Alternate exterior angles measure $(6x - 14)^\circ$ and $(4x + 26)^\circ$. After finding x , find the interior angle at one crossing that forms a linear pair with the alternate exterior angle there.

- | | |
|-----------------|----------------|
| (A) 106 degrees | (B) 74 degrees |
| (C) 20 degrees | (D) 53 degrees |

21. A transversal cuts parallel lines. One co-interior angle is 8° more than three times the other co-interior angle. Find the LARGER co-interior angle.

- | | |
|-----------------|-----------------|
| (A) 43 degrees | (B) 129 degrees |
| (C) 137 degrees | (D) 94 degrees |

22. Parallel lines l and m are cut by transversal t at P and Q . At P an angle is $(3x + 25)^\circ$; at Q an angle is $(5x - 5)^\circ$. These two are alternate exterior angles. After finding x , find the supplement of that angle.

- | | |
|----------------|-----------------|
| (A) 70 degrees | (B) 110 degrees |
| (C) 15 degrees | (D) 55 degrees |

23. A transversal cuts parallel lines. One alternate interior angle and the co-interior angle adjacent to it (forming a linear pair at the same vertex) are in the ratio $4 : 5$. Find the alternate interior angle.

- | | |
|-----------------|----------------|
| (A) 100 degrees | (B) 90 degrees |
| (C) 45 degrees | (D) 80 degrees |

24. Parallel lines l and m are cut by a transversal. A pair of corresponding angles measure $(2x + 35)^\circ$ and $(4x + 5)^\circ$. After finding x , find the difference between that corresponding angle and HALF of its co-interior partner.

- | | |
|-----------------|------------------|
| (A) 7.5 degrees | (B) 57.5 degrees |
| (C) 55 degrees | (D) 82.5 degrees |

25. A transversal cuts parallel lines m and n . The corresponding angle at m is $(x + 50)^\circ$ and at n is $(3x - 10)^\circ$. After finding x , find the angle vertically opposite the co-interior partner of that corresponding angle.

- | | |
|-----------------|----------------|
| (A) 80 degrees | (B) 30 degrees |
| (C) 100 degrees | (D) 40 degrees |

26. Parallel lines are cut by a transversal so that two co-interior angles are $(3x + 15)^\circ$ and $(2x + 25)^\circ$. After finding x , find the exterior linear-pair partner of the LARGER co-interior angle.

- | | |
|----------------|----------------|
| (A) 99 degrees | (B) 28 degrees |
| (C) 81 degrees | (D) 96 degrees |

27. A transversal cuts parallel lines. An alternate interior angle equals $(5x - 18)^\circ$ and its co-interior partner at the same vertex equals $(4x + 9)^\circ$. After finding x , find the alternate interior angle.

(A) 87 degrees

(B) 93 degrees

(C) 21 degrees

(D) 78 degrees

28. Parallel lines l and m are cut by a transversal. A corresponding angle is $(8x + 4)^\circ$ at l and $(6x + 22)^\circ$ at m . After finding x , a ray bisects the corresponding angle at l , then a second ray bisects one of those halves. Find the angle the second ray makes with the nearer arm.

(A) 19 degrees

(B) 38 degrees

(C) 76 degrees

(D) 9.5 degrees

29. A transversal cuts parallel lines. The eight angles take just two values, the larger exceeding the smaller by 44° . Find the SMALLER value.

(A) 112 degrees

(B) 68 degrees

(C) 44 degrees

(D) 90 degrees

30. Two lines intersect at O so that angle $AOC = (9x)^\circ$ and the adjacent angle $COB = (3x)^\circ$ form a linear pair. Ray OD bisects angle AOC and ray OE bisects angle COB . Find angle DOE .

(A) 67.5 degrees

(B) 45 degrees

(C) 135 degrees

(D) 90 degrees

31. Parallel lines are cut by a transversal. The alternate interior angle is $(2x + 9)^\circ$ and the alternate interior angle on the other line is $(3x - 6)^\circ$. After finding x , find the co-interior partner of that angle at the same crossing.

(A) 141 degrees

(B) 39 degrees

(C) 15 degrees

(D) 51 degrees

32. A transversal cuts parallel lines. Co-interior angles are $(x + 24)^\circ$ and $(2x)^\circ$. After finding x , find the alternate exterior angle of the SMALLER co-interior angle.

(A) 104 degrees

(B) 76 degrees

(C) 52 degrees

(D) 38 degrees

33. Two lines intersect at O . Angle $AOC = (10x + 5)^\circ$ and its vertically opposite angle $= (8x + 23)^\circ$. After finding the angle, ray OF bisects the linear-pair neighbour of angle AOC . Find the angle between OF and the arm shared with angle AOC .

(A) 95 degrees

(B) 85 degrees

(C) 42.5 degrees

(D) 47.5 degrees

34. Parallel lines l and m are cut by a transversal. $\angle A$ on l and $\angle B$ on m are corresponding angles with $\angle A = (3 \cdot \angle B - 100)^\circ$. Find $\angle A$.

(A) 150 degrees

(B) 50 degrees

(C) 75 degrees

(D) 100 degrees

35. Segment AB is parallel to segment CD . Point E lies between the two parallels, joined to B on AB and to C on CD , forming a Z-bend at E . $\angle ABE = 48^\circ$ and $\angle DCE = 71^\circ$. Find $\angle BEC$.

- (A) 61 degrees
(C) 119 degrees

- (B) 23 degrees
(D) 133 degrees

36. Segment AB is parallel to segment CD, with point E joined to B and to C. A helper line through E parallel to both shows that $\angle ABE = 62^\circ$ and $\angle DCE = 27^\circ$ fall on the SAME side of that helper line (a U-bend, not a Z-bend). Find $\angle BEC$.

- (A) 89 degrees
(C) 55 degrees

- (B) 35 degrees
(D) 118 degrees

37. Segment AB is parallel to segment CD with point E between them, joined to B and C in a Z-bend. $\angle ABE = 33^\circ$ and the bend $\angle BEC = 118^\circ$. Using a helper line through E parallel to both, find $\angle DCE$.

- (A) 85 degrees
(C) 62 degrees

- (B) 151 degrees
(D) 47 degrees

38. Segment PQ is parallel to segment RS. Point T lies between them in a Z-bend joined to Q and R. $\angle PQT = (4x)^\circ$ and $\angle SRT = (3x)^\circ$, and the bend $\angle QTR = 105^\circ$. Find the larger of $\angle PQT$ and $\angle SRT$.

- (A) 45 degrees
(C) 75 degrees

- (B) 60 degrees
(D) 105 degrees

39. Segment AB $\parallel$ segment CD. A point E between them is joined to B and C; the bend at E is $\angle BEC = 90^\circ$. The helper line through E shows $\angle ABE = (2x + 5)^\circ$ and $\angle DCE = (3x - 5)^\circ$ as the two Z-bend pieces. Find $\angle DCE$.

- (A) 49 degrees
(C) 54 degrees

- (B) 41 degrees
(D) 36 degrees

40. Three parallel lines l, m, n (in order) are cut by one transversal. At l an interior angle of 68° opens toward the transversal on the right. Find the interior angle at n on the LEFT side of the transversal.

- (A) 68 degrees
(C) 112 degrees

- (B) 22 degrees
(D) 44 degrees

41. Segment AB $\parallel$ segment EF, and BG is drawn from B to a point G on EF. $\angle ABG = 55^\circ$. Using that AB $\parallel$ EF and BG is a transversal, find $\angle BGF$ (the angle at G on the F-side, alternate interior to $\angle ABG$).

- (A) 125 degrees
(C) 90 degrees

- (B) 35 degrees
(D) 55 degrees

42. A Z-bend between parallels AB and CD gives $\angle ABE = 39^\circ$ and $\angle DCE = 51^\circ$ as the two pieces of $\angle BEC$. Find $\angle BEC$ and then its supplement.

- (A) the bend is 90° and its supplement is 180°
(B) the bend is 12° and its supplement is 168°

(C) the bend is 90° and its supplement is 90°

(D) the bend is 90° and its supplement is 0°

43. Parallel lines l and m are cut by a transversal. The bisectors of the two co-interior angles are drawn. The angle between the two bisectors is:

(A) 45 degrees

(B) 180 degrees

(C) it depends on the angles

(D) 90 degrees

44. Parallel lines l and m are cut by a transversal. A corresponding angle measures 96° . The bisector of the corresponding angle at l and the bisector of the corresponding angle at m are drawn. What is the relationship between these two bisectors?

(A) they are parallel

(B) they are perpendicular

(C) they meet at 96°

(D) they meet at 48°

45. Parallel lines l and m are cut by a transversal. A pair of alternate interior angles each measure 110° . The bisector of one and the bisector of the other are drawn. The angle between the two bisectors is:

(A) 0 degrees (the bisectors are parallel)

(B) 55 degrees

(C) 110 degrees

(D) 90 degrees

46. Parallel lines l and m are cut by a transversal. An interior angle at l is 74° ; its co-interior partner $\angle Q$ at m is bisected. Find the acute angle the bisector of $\angle Q$ makes with the transversal.

(A) 37 degrees

(B) 74 degrees

(C) 106 degrees

(D) 53 degrees

47. Two lines intersect at O with $\angle AOC = (5x)^\circ$ and adjacent $\angle COB = (x)^\circ$ forming a linear pair. Ray OD bisects $\angle AOC$. Find $\angle DOB$.

(A) 90 degrees

(B) 75 degrees

(C) 105 degrees

(D) 15 degrees

48. A transversal cuts parallel lines so that one of the eight angles is $(3x + 18)^\circ$ and the angle vertically opposite to it is $(5x - 24)^\circ$. After finding x , find the co-interior partner of that angle.

(A) 81 degrees

(B) 21 degrees

(C) 99 degrees

(D) 162 degrees

49. Parallel lines l and m are cut by transversal t . $\angle 1$ and $\angle 2$ are alternate interior angles; $\angle 2$ corresponds to $\angle 3$ across a second crossing; $\angle 3$ forms a linear pair with $\angle 4$. If $\angle 1 = 47^\circ$, find $\angle 4$.

(A) 133 degrees

(B) 47 degrees

(C) 43 degrees

(D) 90 degrees

50. Parallel lines l and m are cut by a transversal. A corresponding angle is 100° ; its co-interior partner is 80° . The bisector of the 100° angle and the bisector of the 80° co-

interior angle are drawn. Find the acute angle between the two bisectors.

- (A) 50 degrees
- (B) 90 degrees
- (C) 40 degrees
- (D) 10 degrees

51. A transversal cuts two lines. A pair of co-interior angles measure 97° and 84° . Decide whether the lines are parallel.

- (A) Parallel; the co-interior angles sum to 180°
- (B) Perpendicular; one angle is near 90°
- (C) Not parallel; the co-interior angles sum to 181° , which is 1° over 180°
- (D) Not parallel; the angles differ by 13°

52. A transversal cuts two lines. A pair of alternate interior angles measure 63° at one line and 63° at the other. What can you conclude?

- (A) The lines are not parallel
- (B) The lines are perpendicular
- (C) Nothing can be concluded
- (D) The lines are parallel

53. A transversal cuts two lines. A pair of corresponding angles measure 78° and 82° . What can you conclude?

- (A) The lines are parallel
- (B) The lines are perpendicular
- (C) The lines are not parallel
- (D) The lines meet at 4°

54. Which single condition does NOT by itself guarantee that two lines cut by a transversal are parallel?

- (A) A pair of alternate interior angles are equal
- (B) A pair of co-interior (same-side interior) angles are equal
- (C) A pair of corresponding angles are equal
- (D) A pair of co-interior angles sum to 180°

55. Which statement is FALSE for two lines cut by a transversal?

- (A) Vertically opposite angles are equal whether or not the lines are parallel
- (B) Corresponding angles are equal whenever the two lines are parallel
- (C) Co-interior angles are supplementary whenever the two lines are parallel
- (D) Alternate interior angles are supplementary whenever the two lines are parallel

56. A transversal cuts two lines. A pair of co-interior angles measure 108° and 72° . What can you conclude?

- (A) The lines are not parallel
- (B) The lines are perpendicular
- (C) Nothing can be decided
- (D) The lines are parallel

57. One transversal cuts two lines making a pair of alternate exterior angles of 119° and 119° . A SECOND transversal cuts the same two lines making corresponding angles of 64° and 70° . From the FULL information, the two lines are:

- (A) not parallel
- (B) parallel
- (C) perpendicular
- (D) cannot be decided

58. Line a is perpendicular to line b, and line c is parallel to line b. A transversal makes a 28° angle with line a. Find the acute angle the same transversal makes with line c.

(A) 62 degrees

(B) 28 degrees

(C) 90 degrees

(D) 118 degrees

59. A transversal crosses line a at exactly 90° . Line b is parallel to line a, and one angle the transversal makes with b is $(7x - 13)^\circ$. Find the value of x.

(A) 13

(B) 90

(C) 11

(D) $103/7$ (about 14.7)

60. Line a $\perp$ line b, and line b $\perp$ line c (all in one plane). A transversal makes 37° with line a. Find the acute angle the SAME transversal makes with line c.

(A) 53 degrees

(B) 90 degrees

(C) 37 degrees

(D) 143 degrees

Solutions — Mathematics (Class 7), Chapter 5: Parallel and Intersecting Lines — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	B	21	C	31	A	41	D	51	C
2	C	12	D	22	B	32	A	42	C	52	D
3	D	13	D	23	D	33	C	43	D	53	C
4	D	14	B	24	A	34	B	44	A	54	B
5	B	15	B	25	C	35	C	45	A	55	D
6	B	16	A	26	C	36	B	46	D	56	D
7	D	17	B	27	A	37	A	47	C	57	A
8	B	18	A	28	A	38	B	48	C	58	A
9	A	19	D	29	B	39	A	49	A	59	D
10	C	20	B	30	D	40	C	50	B	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) A linear pair sums to 180, so $(3x+14)+(2x+16)=180$, giving $5x+30=180$ and $x=30$. Then angle $AOC = 3(30)+14 = 104$. Its bisector OE splits it into two equal halves, so angle $EOC = 104/2 = 52$ degrees. Reporting the whole angle 104 ignores the bisector; 38 bisects the adjacent angle (76); 26 halves twice. Angle $EOC = 52$ degrees.

2. (C) Vertically opposite angles are equal: $4x-6 = 2x+30$, so $2x = 36$ and $x = 18$. The angle itself is $4(18)-6 = 66$. Its linear-pair neighbour is its supplement, $180 - 66 = 114$ degrees. Stopping at the vertical angle gives 66; 18 reports x ; 33 halves the original. The neighbour is 114 degrees.

3. (D) Opposite angles are equal and the four total 360: $2(2x)+2(x+30)=360$, so $6x+60=360$ and $x=50$. The two distinct angles are $2x=100$ and $x+30=80$. OF bisects a 100° angle into two equal halves, so the angle to the nearer arm is $100/2 = 50$ degrees. Choosing 100 forgets the bisector; 40 halves the 80° angle; 80 reports the other angle. Angle to the nearer arm is 50 degrees.

4. (D) Linear pair: $(5x+8)+(4x+10)=180$, so $9x+18=180$ and $x=18$. The angles are $5(18)+8 = 98$ and $4(18)+10 = 82$. The larger is 98; half the smaller is $82/2 = 41$; the difference is $98 - 41 = 57$ degrees. Taking the difference of the two angles gives 16; 98 stops too early; 41 reports the half itself. The difference is 57 degrees.

5. (B) Vertically opposite angles are equal, so $7x-5 = 3x+47$, giving $4x = 52$ and $x = 13$. Choosing 23 mis-handles the constants; 26 doubles x ; 6.5 halves it. $x = 13$.

6. (B) Angles around a point sum to 360: $2y + (3y+20) + (4y-20) = 360$, so $9y = 360$ and $y = 40$. The angles are 80, 140, and 140. The largest is 140, whose supplement is $180 - 140 = 40$ degrees. Choosing 140 forgets to supplement; 100 supplements the smallest; 20 mis-subtracts. The supplement of the largest is 40 degrees.

7. (D) Corresponding angles on parallels are equal: $5x-12 = 3x+8$, so $2x = 20$ and $x = 10$. The corresponding angle is $3(10)+8 = 38$. Its bisector splits it into two halves of 19, so the acute angle with the transversal is 19 degrees. Choosing 38 forgets the bisector; 71 bisects the supplement; 76 doubles. The bisector makes 19 degrees.

8. (B) Alternate interior angles are equal: $3x-5 = x+41$, so $2x = 46$ and $x = 23$. The angle is $3(23)-5 = 64$. Its co-interior partner at the same crossing forms a linear pair, so it is $180 - 64 = 116$ degrees. Stopping at the angle itself gives 64; 23 reports x ; 46 doubles x . The co-interior partner is 116 degrees.

9. (A) Angle BOD is vertically opposite angle AOC, so angle BOD = 70. Its bisector OE splits it into two 35° halves, so the angle between OE and OB is 35 degrees. Choosing 70 forgets the bisector; 110 is the supplement; 55 mis-halves. Angle EOB = 35 degrees.

10. (C) Vertically opposite: $6x+4 = 8x-30$, so $2x = 34$ and $x = 17$. Angle AOC = $6(17)+4 = 106$. Its linear-pair neighbour is $180 - 106 = 74$ degrees. Choosing 106 stops early; 17 reports x ; 53 halves. The neighbour is 74 degrees.

11. (B) The four angles sum to 360: $4a + (18+36+54) = 360$, so $4a + 108 = 360$, $4a = 252$ and $a = 63$. The angles are 63, 81, 99, 117. Largest minus smallest is $117 - 63 = 54$ degrees. Choosing 18 takes one gap; 36 takes two gaps; 108 is the constant total. The difference is 54 degrees.

12. (D) The ratio parts total $2+3+5 = 10$, filling 360, so one part is 36. The angles are 72, 108, and 180. The largest is the straight angle 180, whose supplement is $180 - 180 = 0$ degrees. Choosing 30, 60, or 90 supplements a smaller angle or mis-totals. The supplement of the largest (180°) is 0 degrees.

13. (D) Sum is 360: $3x + (2x+10) + (x+20) + (4x-30) = 360$, so $10x + 0 = 360$ and $x = 36$. The four angles are 108, 82, 56, 114. The largest is 114, half of which is 57 degrees. Choosing 114 forgets to halve; 54 halves 108; 36 reports x . Half the largest is 57 degrees.

14. (B) The two $(2x+10)^\circ$ marks are vertically opposite, and $(3x)^\circ$ is adjacent, forming a linear pair: $(2x+10)+(3x)=180$, so $5x+10=180$, $5x=170$ and $x=34$. Then $3x = 102$ and its supplement is $180 - 102 = 78$ degrees. Choosing 102 forgets the supplement; 60 and 120 come from mis-solving. The supplement is 78 degrees.

15. (B) Co-interior angles are supplementary: $(4x+10)+(5x-1)=180$, so $9x+9=180$, $9x=171$ and $x=19$. The two are $4(19)+10 = 86$ and $5(19)-1 = 94$. The larger is 94, and its alternate interior angle equals it (alternate interior angles are equal on parallels), so 94 degrees. Choosing 86 picks the smaller; 90 assumes perpendicular; 85 mis-computes. The answer is 94 degrees.

16. (A) Corresponding angles are equal: $7x-20 = 4x+16$, so $3x = 36$ and $x = 12$. The corresponding angle is $4(12)+16 = 64$. Its co-interior partner (same-side interior at the other crossing) is supplementary: $180 - 64 = 116$ degrees. Choosing 64 stops at the corresponding angle; 12 reports x ; 44 mis-computes. The co-interior partner is 116 degrees.

17. (B) The 137° exterior angle has an interior linear-pair partner of $180 - 137 = 43$ at the upper line. On parallels, the co-interior partner of that 43° interior angle at the lower line is its supplement, $180 - 43 = 137$ degrees. Choosing 43 stops at the upper interior angle; 90 assumes perpendicular; 47 mis-computes. The required lower-line angle is 137 degrees.

18. (A) Five angles around a point sum to 360: $y+2y+3y+4y+5y = 15y = 360$, so $y = 24$. The angles are 24, 48, 72, 96, 120. The largest is 120; the second-smallest is 48, whose supplement is $180 - 48 = 132$. The difference is $132 - 120 = 12$ degrees. Choosing 108 mis-supplements; 132 stops at the supplement; 60 mis-pairs. The difference is 12 degrees.

19. (D) Corresponding angles are equal, so angle 2 = 121. Vertically opposite, so angle 3 = 121. Its linear-pair neighbour angle 4 = $180 - 121 = 59$ degrees. Stopping early gives 121; 90 assumes perpendicular; 29 over-subtracts. Angle 4 = 59 degrees.

20. (B) Alternate exterior angles are equal: $6x-14 = 4x+26$, so $2x = 40$ and $x = 20$. Each alternate exterior angle is $6(20)-14 = 106$. The interior angle forming a linear pair with it is $180 - 106 = 74$ degrees. Choosing 106 stops at the exterior angle; 20 reports x ; 53 halves. The linear-pair interior angle is 74 degrees.

21. (C) Let the smaller co-interior angle be s ; the larger is $3s + 8$. Co-interior angles are supplementary: $s + (3s + 8) = 180$, so $4s = 172$ and $s = 43$. The larger is $3(43)+8 = 137$ degrees. Choosing 43 reports the smaller; 129 is $3s$ without the $+8$; 94 mis-adds. The larger co-interior angle is 137 degrees.

22. (B) Alternate exterior angles are equal: $3x+25 = 5x-5$, so $2x = 30$ and $x = 15$. The angle is $3(15)+25 = 70$. Its supplement is $180 - 70 = 110$ degrees. Choosing 70 forgets to supplement; 15 reports x ; 55 halves the supplement. The supplement is 110 degrees.

- 23. (D)** The alternate interior angle and its co-interior neighbour at the same vertex form a linear pair, summing to 180, in ratio 4:5. The parts total 9, so one part is 20. The alternate interior angle is $4(20) = 80$ degrees and the co-interior is $5(20) = 100$. Choosing 100 picks the co-interior; 90 assumes equal halves; 45 mis-divides. The alternate interior angle is 80 degrees.
- 24. (A)** Corresponding angles are equal: $2x+35 = 4x+5$, so $2x = 30$ and $x = 15$. The angle is $2(15)+35 = 65$. Its co-interior partner is $180 - 65 = 115$, and half of that is 57.5. The difference is $65 - 57.5 = 7.5$ degrees. Choosing 57.5 stops at the half; 55 halves 110; 82.5 adds. The difference is 7.5 degrees.
- 25. (C)** Corresponding angles are equal: $x+50 = 3x-10$, so $2x = 60$ and $x = 30$. The angle is $30+50 = 80$. Its co-interior partner is $180 - 80 = 100$, and the angle vertically opposite that partner equals it, so 100 degrees. Choosing 80 stops at the corresponding angle; 30 reports x ; 40 halves. The answer is 100 degrees.
- 26. (C)** Co-interior angles are supplementary: $(3x+15)+(2x+25)=180$, so $5x+40=180$, $5x=140$ and $x=28$. The two are $3(28)+15 = 99$ and $2(28)+25 = 81$. The larger co-interior is 99; its exterior linear-pair partner is $180 - 99 = 81$ degrees. Choosing 99 stops at the larger co-interior; 28 reports x ; 96 mis-computes. The exterior partner is 81 degrees.
- 27. (A)** The alternate interior angle and its co-interior partner at the same vertex form a linear pair: $(5x-18)+(4x+9)=180$, so $9x-9=180$, $9x=189$ and $x=21$. The alternate interior angle is $5(21)-18 = 87$ degrees, and its co-interior partner is $4(21)+9 = 93$, with $87+93 = 180$. Choosing 93 picks the co-interior partner; 21 reports x ; 78 mis-computes. The alternate interior angle is 87 degrees.
- 28. (A)** Corresponding angles are equal: $8x+4 = 6x+22$, so $2x = 18$ and $x = 9$. The angle is $8(9)+4 = 76$. The first bisector makes two halves of 38; the second ray bisects one 38° half into two 19° pieces, so the angle to the nearer arm is 19 degrees. Choosing 76 forgets bisecting; 38 stops after one bisection; 9.5 over-halves. The second ray makes 19 degrees.
- 29. (B)** The two values are supplementary (sum 180) and differ by 44. So smaller + larger = 180 and larger - smaller = 44; adding, $2(\text{larger}) = 224$, larger = 112 and smaller = 68 degrees. Choosing 112 picks the larger; 44 is the difference; 90 assumes equal. The smaller value is 68 degrees.
- 30. (D)** Linear pair: $9x + 3x = 180$, so $12x = 180$ and $x = 15$. Then angle AOC = 135 and angle COB = 45. OD bisects AOC into 67.5° and OE bisects COB into 22.5° , both measured from the common arm OC, so angle DOE = $67.5 + 22.5 = 90$ degrees. (The bisectors of two adjacent supplementary angles are always perpendicular.) Choosing 67.5 takes one half; 45 reports COB; 135 reports AOC. Angle DOE = 90 degrees.
- 31. (A)** Alternate interior angles are equal: $2x+9 = 3x-6$, so $x = 15$. The angle is $2(15)+9 = 39$. Its co-interior partner is $180 - 39 = 141$ degrees. Choosing 39 stops at the angle itself;

15 reports x ; 51 mis-computes. The co-interior partner is 141 degrees.

32. (A) Co-interior angles are supplementary: $(x+24)+(2x)=180$, so $3x+24=180$, $3x=156$ and $x=52$. The two are $x+24 = 76$ and $2x = 104$; the smaller is 76. The smaller co-interior angle's exterior linear-pair partner is $180 - 76 = 104$, and the alternate exterior angle equals that exterior angle, so 104 degrees. Choosing 76 reports the smaller co-interior; 52 reports x ; 38 halves. The alternate exterior of the smaller co-interior is 104 degrees.

33. (C) Vertically opposite: $10x+5 = 8x+23$, so $2x = 18$ and $x = 9$. Angle AOC = $10(9)+5 = 95$. Its linear-pair neighbour is $180 - 95 = 85$. OF bisects that 85° neighbour into two 42.5° halves, so the angle between OF and the shared arm is 42.5 degrees. Choosing 95 reports AOC; 85 the neighbour; 47.5 halves AOC. The answer is 42.5 degrees.

34. (B) Corresponding angles on parallels are equal, so $\angle A = \angle B$. Substituting gives $\angle A = 3\angle A - 100$, so $2\angle A = 100$ and $\angle A = 50$ degrees. Choosing 150 forgets they are equal; 75 mis-solves; 100 reports the constant. $\angle A = 50$ degrees.

35. (C) Draw a helper line through E parallel to both AB and CD. The Z-bend splits $\angle BEC$ into two alternate-interior pieces: one equals $\angle ABE = 48^\circ$ and the other equals $\angle DCE = 71^\circ$. So $\angle BEC = 48 + 71 = 119$ degrees. Choosing 61 mis-subtracts; 23 takes the difference; 133 mis-adds. $\angle BEC = 119$ degrees.

36. (B) With the helper line through E, each given angle equals the angle E makes with the helper line by alternate interior angles, but because both fall on the SAME side they SUBTRACT: $\angle BEC = 62 - 27 = 35$ degrees. Choosing 89 adds them as in a Z-bend; 55 mis-computes; 118 supplements. $\angle BEC = 35$ degrees.

37. (A) The helper line splits $\angle BEC$ into two pieces, one equal to $\angle ABE = 33^\circ$ (alternate interior) and the other equal to $\angle DCE$. So $\angle DCE = 118 - 33 = 85$ degrees. Choosing 151 adds; 62 supplements 118; 47 mis-computes. $\angle DCE = 85$ degrees.

38. (B) In a Z-bend the helper line gives $\angle QTR = \angle PQT + \angle SRT$, so $4x + 3x = 105$, $7x = 105$ and $x = 15$. Then $\angle PQT = 60$ and $\angle SRT = 45$. The larger is 60 degrees. Choosing 45 picks the smaller; 75 mis-solves; 105 reports the bend. The larger is 60 degrees.

39. (A) Z-bend: $\angle ABE + \angle DCE = \angle BEC$, so $(2x+5)+(3x-5)=90$, $5x=90$ and $x=18$. Then $\angle DCE = 3(18)-5 = 49$ degrees and $\angle ABE = 41$. Choosing 41 picks $\angle ABE$; 54 forgets the -5 ; 36 mis-computes. $\angle DCE = 49$ degrees.

40. (C) The 68° interior angle at l corresponds to the matching 68° interior angle at n on the SAME (right) side. The interior angle at n on the LEFT side is its linear-pair neighbour, $180 - 68 = 112$ degrees. Choosing 68 forgets the side switch; 22 over-subtracts; 44 mis-computes. The left-side interior angle at n is 112 degrees.

41. (D) $AB \parallel EF$ and BG is a transversal, so $\angle ABG$ and $\angle BGF$ are alternate interior angles and are equal: $\angle BGF = 55$ degrees. Choosing 125 supplements; 35 takes the complement; 90 assumes perpendicular. $\angle BGF = 55$ degrees.

42. (C) The Z-bend gives $\angle BEC = 39 + 51 = 90$ degrees. Its supplement is $180 - 90 = 90$ degrees, since 90° is self-supplementary. The option giving supplement 180 doubles; the 12° option subtracts instead of adding; the 0° option mis-supplements. The bend is 90° and its supplement is 90° .

43. (D) Two co-interior angles sum to 180, so their halves sum to 90. The bisectors of co-interior angles between parallel lines are therefore always perpendicular; the angle between them is 90 degrees. Choosing 45 halves once more; 180 forgets the figure; 'it depends' is false because the co-interior sum is always 180. The angle between the bisectors is 90 degrees.

44. (A) Corresponding angles at l and m are equal (both 96°) and similarly oriented. Bisecting each gives a ray making 48° with the transversal in the same direction, so the two bisectors are themselves corresponding rays for the transversal cutting two parallel lines — hence parallel. Choosing perpendicular or a meeting angle ignores that equal corresponding angles bisect into parallel rays. The bisectors are parallel.

45. (A) Alternate interior angles are equal (both 110°) but oppositely oriented across the transversal. Bisecting each makes 55° with the transversal on alternate sides, so the two bisector rays are themselves alternate interior for the parallels and are parallel; the angle between them is 0 degrees. Choosing 55 halves once; 110 forgets bisecting; 90 assumes perpendicular. The bisectors are parallel.

46. (D) The co-interior partner $\angle Q = 180 - 74 = 106$. Bisecting gives $106/2 = 53$, so the acute angle the bisector makes with the transversal is 53 degrees. Choosing 37 bisects the 74° angle; 74 forgets the co-interior step; 106 forgets bisecting. The bisector makes 53 degrees.

47. (C) Linear pair: $5x + x = 180$, so $6x = 180$ and $x = 30$. Then $\angle AOC = 150$ and $\angle COB = 30$. OD bisects $\angle AOC$, so $\angle DOC = 75$. Then $\angle DOB = \angle DOC + \angle COB = 75 + 30 = 105$ degrees. Choosing 90 is the bisector-to-bisector angle, not asked; 75 stops at $\angle DOC$; 15 halves $\angle COB$. $\angle DOB = 105$ degrees.

48. (C) Vertically opposite angles are equal: $3x+18 = 5x-24$, so $2x = 42$ and $x = 21$. The angle is $3(21)+18 = 81$. Its co-interior partner on the parallels is supplementary, $180 - 81 = 99$ degrees. Choosing 81 stops at the angle itself; 21 reports x ; 162 doubles the angle. The co-interior partner is 99 degrees.

49. (A) Alternate interior angles are equal, so $\angle 2 = 47$. Corresponding angles are equal, so $\angle 3 = 47$. Its linear-pair neighbour $\angle 4 = 180 - 47 = 133$ degrees. Stopping early gives 47; 43 takes the complement; 90 assumes perpendicular. $\angle 4 = 133$ degrees.

50. (B) The corresponding angle 100° and its co-interior partner 80° are supplementary ($100+80=180$). The bisector of the 100° makes 50° with the transversal; the bisector of the 80° makes 40° on the same side. Since the parents are co-interior (supplementary), the bisectors are perpendicular: $50 + 40 = 90$ degrees. Choosing 50 or 40 reports a single half; 10 takes the difference. The angle between the bisectors is 90 degrees.

51. (C) Co-interior angles between parallel lines must sum to exactly 180° . Here $97 + 84 = 181$, which is 1° too much, so the lines are NOT parallel. The 'parallel' option mis-states the sum; the difference-of-13 reasoning uses the wrong test; perpendicular is irrelevant. They are not parallel because the co-interior sum is 181° , 1° over 180° .

52. (D) Equal alternate interior angles are exactly the condition that guarantees two lines cut by a transversal are parallel. Since both are 63° , the lines are parallel. The other options ignore this converse test. The lines are parallel.

53. (C) Corresponding angles are equal only when the lines are parallel. Here $78 \neq 82$, so the lines are not parallel. The '4°' option wrongly treats the difference as a crossing angle; perpendicular is irrelevant. The lines are not parallel.

54. (B) For parallelism, co-interior angles must be SUPPLEMENTARY (sum 180°), not equal; equal co-interior angles do not in general prove parallelism. Equal alternate interior angles, equal corresponding angles, and co-interior summing to 180° each guarantee parallelism. So the equal-co-interior condition is the one that does not by itself guarantee parallel lines.

55. (D) On parallel lines, alternate interior angles are EQUAL, not supplementary, so that statement is false. Vertically opposite angles are always equal; corresponding angles are equal on parallels; co-interior angles are supplementary on parallels — all true. The false statement claims alternate interior angles are supplementary.

56. (D) Co-interior angles between parallels sum to 180° . Here $108 + 72 = 180$ exactly, which is the parallel condition, so the lines are parallel. The 'not parallel' and 'nothing' options ignore the satisfied test; perpendicular is irrelevant. The lines are parallel.

57. (A) The first transversal's equal alternate exterior angles suggest parallel, but the second transversal gives unequal corresponding angles ($64 \neq 70$). Genuinely parallel lines give equal corresponding angles for EVERY transversal, so the second measurement contradicts parallelism. The lines are not parallel; the first pair was coincidental. The lines are not parallel.

58. (A) Since $c \parallel b$ and $a \perp b$, line a is also perpendicular to c . The transversal makes 28° with a ; relative to c (perpendicular to a) it makes $90 - 28 = 62$ degrees (acute). Choosing 28 forgets the perpendicular turn; 90 reports the perpendicularity; 118 is the obtuse partner. The acute angle with c is 62 degrees.

59. (D) If $b \parallel a$ and the transversal is perpendicular to a , it is perpendicular to b too, so every angle it makes with b is 90° . Setting $7x - 13 = 90$ gives $7x = 103$ and $x = 103/7$, about 14.7. Choosing 13 ignores the equation; 90 confuses x with the angle; 11 mis-solves. $x = 103/7$ (about 14.7).

60. (C) If $a \perp b$ and $b \perp c$, then a and c are parallel (both perpendicular to b). A transversal makes equal acute angles with parallel lines, so the angle with c equals the angle with a , namely 37 degrees. Choosing 53 turns through a perpendicular incorrectly; 90 reports the perpendicularity; 143 is obtuse. The acute angle with c is 37 degrees.